Lecture 27.06: Correlation Analysis & Multicollinearity Diagnostics

📌 Executive Summary

Pearson correlation coefficient formulation, correlation matrix heatmaps, identifying primary risk predictors for loan approval, and diagnosing feature collinearity.

🏛️ Theoretical Formulation & Mathematical Architecture

📈 Pearson Product-Moment Correlation Formulation

The linear association between two continuous random variables X and Y across a sample of m observations is measured by the **Pearson Correlation Coefficient** r_{xy}:

$$\boxed{ r_{xy} = \frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^m (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^m (y_i - \bar{y})^2}} = \frac{ \text{Cov}(X, Y)}{\sigma_X \sigma_Y} }$$

Where:

- $r_{xy} \in [-1.0, +1.0]$.
- $r_{xy} = +1.0$ indicates perfect deterministic positive collinearity.
- $r_{xy} = -1.0$ indicates perfect deterministic inverse collinearity.
- $r_{xy} = 0.0$ signifies the complete absence of any linear dependency.

🔍 Credit Risk Correlation Ranking Hierarchy

In our encoded SecureTrust Bank design matrix, sorting features by their absolute correlation with the supervisory label `Loan_Approved` yields critical domain insights:

```
flowchart LR
    A["Loan_Approved Target"] --> B["High Positive Correlation<br/>(+0.30 to +0.60
    A --> C["High Negative Correlation<br/>(-0.25 to -0.55)"]
    A --> D["Weak / Neutral Correlation<br/>(-0.05 to +0.05)"]

    B --> B1["Credit_Score (Primary Underwriting Pillar)"]
```

```
    B --> B2["Savings & Collateral_Value (Capital Buffer)"]

    C --> C1["DTI_Ratio (Excess Debt Burden)"]
    C --> C2["Existing_Loans (Leverage Constraint)"]

    D --> D1["Demographic Features (Gender, Marital_Status)"]
```

Multicollinearity Alert:

Features displaying severe pairwise correlation ($|r_{jk}| > 0.85$) inflate the parameter standard errors in linear models. In baseline loan data, `Applicant_Income` and `Total_Income` will naturally exhibit strong covariance, warranting careful ratio-based feature engineering.

💻 Vectorized Python Implementation

```
import pandas as pd

# Load and compute correlation against target
# Assume df_encoded is ready
corr_series = df_encoded.corr()["Loan_Approved"].sort_values(ascending=False)
print("--- Features Positively Correlated with Loan Approval ---")
print(corr_series[corr_series > 0].head(5))

print("
--- Features Negatively Correlated with Loan Approval ---")
print(corr_series[corr_series < 0].tail(5))
```

🔬 Key Engineering Takeaways & Invariants

1. **Mathematical Grounding**: Every transformation and classification step directly addresses the underlying empirical distribution and financial risk constraints.
2. **Reproducibility**: Clean, leak-free pipeline partitioning guarantees that performance metrics match production operating realities.
3. **Domain Alignment**: Aligning model evaluation with the asymmetric cost of financial errors protects institutional solvency.